

GU/CSE241/A2

Exercise Solutions

Demo Recordings Exercise 1 - Basic Password Recovery

```
(kali@DESKTOP-K00BQCF) - [/mnt/.../Security/assignments/2/E1]
$ ls
total 0

(kali@DESKTOP-K00BQCF) - [/mnt/.../Security/assignments/2/E1]
$ micro solve.sh

(kali@DESKTOP-K00BQCF) - [/mnt/.../Security/assignments/2/E1]
$ ./solve.sh
$1$aDuXb57Z$m57SlJ7zppV9jF07Jl/8p0
stat: ./hashes: No such file or directory

(kali@DESKTOP-K00BQCF) - [/mnt/.../Security/assignments/2/E1]
$ ls
total 0
-rwxrwxrwx 1 kali kali 274 Nov  7 15:33 solve.sh

(kali@DESKTOP-K00BQCF) - [/mnt/.../Security/assignments/2/E1]
$ micro ./solve.sh

(kali@DESKTOP-K00BQCF) - [/mnt/.../Security/assignments/2/E1]
$ ./solve.sh
Warning: detected hash type "md5crypt", but the string is also recognized as "md5crypt-long"
Use the "--format=md5crypt-long" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (md5crypt, crypt(3) $1$ (and variants) [MD5 256/256 AVX2 8x3])
Will run 12 OpenMP threads
Proceeding with single, rules:Single
Press Ctrl-C to abort, or send SIGUSR1 to john process for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
password (??)
1g 0:00:00:00 DONE 2/3 (2025-11-07 15:33) 2.564g/s 2953p/s 2953c/s 2953C/s 123456..88888888
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@DESKTOP-K00BQCF) - [/mnt/.../Security/assignments/2/E1]
$ #done
```



Exercise 2.1 - Complex Password Analysis -> INCREMENTAL

```

0g 0:00:02:37 0g/s 442269p/s 442269c/s 442269C/s madjhs..masivy
0g 0:00:02:38 0g/s 442772p/s 442772c/s 442772C/s lsvvta..lsjsws
0g 0:00:02:41 0g/s 443308p/s 443308c/s 443308C/s stmyrubs..stmewhbe
0g 0:00:02:42 0g/s 443835p/s 443835c/s 443835C/s briserri..briecich
0g 0:00:03:19 0g/s 449977p/s 449977c/s 449977C/s casadl..catncd
0g 0:00:03:30 0g/s 444421p/s 444421c/s 444421C/s paoponz..padfrmi
Session aborted

(kali@DESKTOP-K00BQCF) - [/mnt/.../assignments/2/E2/P2]
$ ls
total 0
-rwxrwxrwx 1 kali kali 35 Nov  7 15:53 hash

(kali@DESKTOP-K00BQCF) - [/mnt/.../assignments/2/E2/P2]
$ john --help | fzf

(kali@DESKTOP-K00BQCF) - [/mnt/.../assignments/2/E2/P2]
$ ls
total 0
-rwxrwxrwx 1 kali kali 35 Nov  7 15:53 hash

(kali@DESKTOP-K00BQCF) - [/mnt/.../assignments/2/E2/P2]
$ john --incremental=Upper ./hash
Warning: detected hash type "md5crypt", but the string is also recognized as "md5crypt-long"
Use the "--format=md5crypt-long" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (md5crypt, crypt(3) $1$ (and variants) [MD5 256/256 AVX2 8x3])
Will run 12 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:02 0g/s 453046p/s 453046c/s 453046C/s RRFANI..RRHLIN
0g 0:00:00:03 0g/s 489974p/s 489974c/s 489974C/s KEENIK..KEIGEN
MMA (7)
1g 0:00:00:04 DONE (2025-11-07 15:57) 0.2192g/s 500715p/s 500715c/s 500715C/s FDJCK..METBB
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@DESKTOP-K00BQCF) - [/mnt/.../assignments/2/E2/P2]
$ #done

```

Exercise 2.2 - Complex Password Analysis -> MASK

```

Press 'q' or Ctrl-C to abort, almost any other key for status
ali (7)
1g 0:00:00:00 DONE (2025-11-07 15:45) 5.263g/s 12126p/s 12126c/s 12126C/s tpe..bco
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@DESKTOP-K00BQCF) - [/mnt/.../S5/Security/assignments/2]
$ openssl passwd -1 "M@hr0s" > md5

(kali@DESKTOP-K00BQCF) - [/mnt/.../S5/Security/assignments/2]
$ john --mask="?u?s?l?l?d?l" ./md5
Warning: detected hash type "md5crypt", but the string is also recognized as "md5crypt-long"
Use the "--format=md5crypt-long" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (md5crypt, crypt(3) $1$ (and variants) [MD5 256/256 AVX2 8x3])
Will run 12 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:47 15.21% (ETA: 15:51:18) 0g/s 487982p/s 487982c/s 487982C/s W}iu7o..0,nu7o
Session aborted

(kali@DESKTOP-K00BQCF) - [/mnt/.../S5/Security/assignments/2]
$ openssl passwd -1 "@l3" > md5

(kali@DESKTOP-K00BQCF) - [/mnt/.../S5/Security/assignments/2]
$ john --mask="?s?l?d" ./md5
Warning: detected hash type "md5crypt", but the string is also recognized as "md5crypt-long"
Use the "--format=md5crypt-long" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (md5crypt, crypt(3) $1$ (and variants) [MD5 256/256 AVX2 8x3])
Will run 12 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
@l3 (7)
1g 0:00:00:00 DONE (2025-11-07 15:47) 5.263g/s 18189p/s 18189c/s 18189C/s ^g2..]a9
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@DESKTOP-K00BQCF) - [/mnt/.../S5/Security/assignments/2]
$ that

```

Exercise 3 - Creating Cusom Rule

```

Warning: detected hash type "md5crypt", but the string is also recognized as "md5crypt-long"
Use the "--format=md5crypt-long" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (md5crypt, crypt(3) $1$ (and variants) [MD5 256/256 AVX2 8x3])
Will run 12 OpenMP threads
No "./mahros.rules" mode rules found in /etc/john/john.conf

(kali@DESKTOP-K00BQCF)-[/mnt/.../Security/assignments/2/E3]
$ john --wordlist=./wordlist --rules=$(pwd)/mahros.rules ./hash
Warning: detected hash type "md5crypt", but the string is also recognized as "md5crypt-long"
Use the "--format=md5crypt-long" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (md5crypt, crypt(3) $1$ (and variants) [MD5 256/256 AVX2 8x3])
Will run 12 OpenMP threads
No "/mnt/d/Workspace/university/S5/Security/assignments/2/E3/mahros.rules" mode rules found in /etc/john/john.conf

(kali@DESKTOP-K00BQCF)-[/mnt/.../Security/assignments/2/E3]
$ cat ./mahros.rules >> /etc/john/john.conf
zsh: permission denied: /etc/john/john.conf

(kali@DESKTOP-K00BQCF)-[/mnt/.../Security/assignments/2/E3]
$ sudo cat ./mahros.rules >> /etc/john/john.conf
zsh: permission denied: /etc/john/john.conf

(kali@DESKTOP-K00BQCF)-[/mnt/.../Security/assignments/2/E3]
$ john --wordlist=./wordlist --rules=mahros ./hash
Warning: detected hash type "md5crypt", but the string is also recognized as "md5crypt-long"
Use the "--format=md5crypt-long" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (md5crypt, crypt(3) $1$ (and variants) [MD5 256/256 AVX2 8x3])
Will run 12 OpenMP threads
No "mahros" mode rules found in /etc/john/john.conf

(kali@DESKTOP-K00BQCF)-[/mnt/.../Security/assignments/2/E3]
$ micro mahros.rules

(kali@DESKTOP-K00BQCF)-[/mnt/.../Security/assignments/2/E3]
$ #done

```



Mahros AL-Qabasy 223106831